

Gabor Splatting for High-Quality Gigapixel Image Representations

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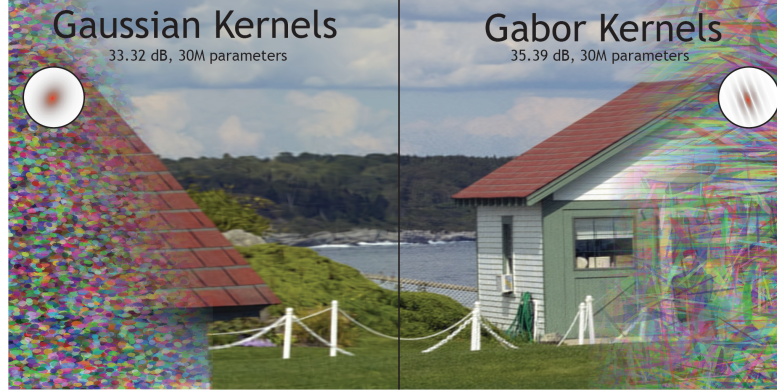


Figure 1: Example of reconstructed gigapixel image with standard Gaussian splatting kernels (left) versus our periodic Gabor splatting kernels (right). Overlaid are the splatting kernels from the trained models to illustrate the shapes of the representative primitive, and we depict a single Gaussian or Gabor kernel in the circle on either side. At the same model size (30M parameters), our approach outperforms Gaussian splatting, evident in detailed regions such as the fence, grass, water, and rocks.

KEYWORDS

Gaussian Splatting, Gigapixel Image, Gabor Filter

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1 ABSTRACT

Differentiable gaussian splatting (3DGS) [Kerbl et al. 2023] has become state of the art for scene representation due to its fast training, high quality results, and ease of editing. The 3DGS model is comprised of a set of isotropic gaussians, each defined with a 3D position, scale, rotation, and spherical harmonics to determine the color. A traditional gaussian rasterization algorithm [Zwicker et al. 2001] is extended to be differentiable, which allows each of the gaussian’s properties to be learned from back-propagation directly,

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without the use of any neural components, supervised from a loss function. By avoiding neural components, the 3DGS representation is multiple times faster to train and infer than the closest state of the art neural radiance field (NeRF) models such as Instant-NGP [Müller et al. 2022]. The high-quality results come at a trade-off, however. To represent scenes accurately, 3DGS introduces millions of gaussians into the scene, which affect training and rendering speed as well as introduce more parameters that occupy VRAM and storage when saved to disc.

In this paper, we specifically focus on learning gigapixel image approximations, which is a useful benchmark for the capabilities of a model to capture intricate details [Müller et al. 2022; Sitzmann et al. 2020]. Toward the goal of reducing the number of gaussians required for an accurate representation, we propose a Gabor splatting (GabSplat) method that uses a Gabor filter as the representation as opposed to a standard gaussian, as shown in Figure 1.

2 OUR APPROACH

Our method uses all the same parameters standard gaussian splatting uses, which defines the gaussian envelope, but also adds a new parameter set for each gaussian defining the periodic element necessary to create the Gabor filter.

Gabor basis. A Gabor filter, shown in Figure 2, is the product of a gaussian envelope with a sinusoidal function:

$$g(x) = \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \cos(fx). \quad (1)$$

We use a cosine wave as the periodic element such that there is a peak at the center of the Gabor. A 2D Gabor takes the form

$$g(x) = \exp\left(-\frac{1}{2}x^T \Sigma^{-1}x\right) \cos(fx_d), \quad (2)$$

where Σ is the covariance matrix for the gaussian envelope and x_d is x in the direction of the periodic effect. Other works use a radial direction to operate on periodically [Chen et al. 2023], but we find this creates a ripple-like effect that hurts performance. Instead of introducing a learnable rotation angle, we use the gaussian’s scaled and rotated x-axis as a fixed direction for periodicity. As the gaussian envelope rotates, so too will the periodic effect, and vice versa. Likewise, the scaling of the envelope will also change the effective frequency of the gabor as well. For example, as the scale of the Gabor becomes smaller, the frequency effectively becomes higher in the global domain, but remains the same relative to the Gabor. We use this strategy to model higher frequencies naturally as the Gabors get smaller during training.

Periodic basis. Directly training a learnable frequency term within each Gabor is challenging due to the periodic nature of the loss landscape for the frequency term. There are many local extrema that are easy for the frequency term to get stuck in, hurting performance. This is a well known phenomena, which is why other tools, such as the Lomb-Scargle periodogram [Zechmeister and Kürster 2009], avoid the issue by using a basis of sinusoidal functions with predetermined frequency terms as opposed to regression on a frequency parameter. In this type of model, the only term being learned is the coefficient for each sinusoid before summing, making the problem linear and easily solvable with regression or as a linear system.

We adapt the filter bank idea to model the periodic part of each Gabor. Explicitly, each Gabor contains a filter bank of F coefficients, corresponding to sinusoidal terms with frequency $f = \{0, \frac{\text{max_freq}}{F-1}, 2\frac{\text{max_freq}}{F-1}, \dots, (F-1)\frac{\text{max_freq}}{F-1}\}$, such that each frequency is a linear interpolation between 0 Hz and max_freq Hz, inclusive. We include 0 Hz such that our model may be equivalent to typical gaussian splatting when all Gabors use 0 Hz only, causing the cosine term to become 1. Thus, evaluating a Gabor at a location x equates to:

$$g(x) = \exp\left(-\frac{1}{2}x^T \Sigma^{-1}x\right) \sum_{i=0}^{F-1} f_i \cos\left(i \frac{\text{max_freq}}{F-1} x_d\right). \quad (3)$$

During training, each f_i has a well-behaved gradient to use for gradient descent.

However, using a full filter bank for training and inference would require a large amount of memory and slow down training. To efficiently train the coefficients in f , we only take the top k coefficients, ranked by absolute value, for computation of Equation 3 during training and inference. This ensures that the terms with the most contribution toward the Gabor are actively being used for the output color, while naturally allowing the top frequency contributors to change based on their coefficient rankings. Additionally, when the model is saved, only those top k frequency coefficients and their frequency need to be stored, since the others are not needed after training. This filter bank style of modelling the Gabor is more VRAM intensive during training, but cheaper to store and load again for inference.

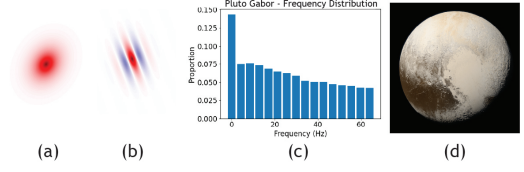


Figure 2: (a) Example of a Gaussian kernel. (b) Example of a Gabor kernel. (c) Frequency distribution of all Gabor kernels in a trained model for (d) the Pluto image .



Figure 3: Qualitative comparison of our approach with Gaussian splatting and I-NGP for some zoomed-in outputs of learned gigapixel image representations. We encourage zooming to show the detail provided with our approach.

3 CONCLUSION AND FUTURE WORK

Across a variety of images in the comparison Figure 3, our Gabor splatting outperforms standard gaussian splatting at similar total parameter counts in peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). We release our code and more results on GitHub at <https://github.com/skywolf829/GaborSplatting>.

To summarize, we propose Gabor Splatting, an alternative for Gaussian splatting which uses a periodic kernel. Our approach, inspired by the compression capabilities offered in the frequency domain, obtains higher quality for the same number of parameters as ordinary gaussian splatting. In the future, we hope to extend this Gabor splatting approach to 3D scenes, increasing the quality of images generated by 3DGS while using fewer parameters.

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